Predicting customer attrition for ShopKeep

OIT 367 – Business Intelligence from Big Data

Final Project

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Note: Will you let us keep your projects? Yes

1. Executive Summary

1.1 Problem description

We were tasked with predicting customer attrition for ShopKeep, a subscription cloud-based point of sale software that aims to replace traditional cash registers. Over the past few years, ShopKeep has seen a sizable portion of its subscribing customers stop using its service after a period of time. The goal of our project is to determine the key factors that determine this customer attrition and to build a model that can predict the likelihood of attrition for current and future customers. To conduct our analysis, we were given six different data sets containing 22 different fields on items such as: account information, daily transaction data, subscription history and revenue contribution. Our job was to build a prediction model that could help ShopKeep's customer relations team to identify which customers they need to focus more of their efforts on retaining. The model would also be helpful in identifying new target customers who are less likely to attrite based on certain customer characteristics.

1.2 Principal findings

Through our analysis, we discovered that several key items have material contributions to customer attrition. The following attributes will tend to decrease customer attrition rates: a) certain business categories; b) larger number of daily transactions; c) number of times deviating from the standard monthly fee; d) higher % of active weeks; and e) increases in recent weekly transaction volume. The following attributes, on the other hand, will tend to increase customer attrition rates: a) certain business categories; b) certain business types; c) higher number of registered licenses; d) number of times receiving a refund; and e) longer subscription time.

1.3 Recommendation

Based on our findings, we can make several recommendations to the company in regards to both customer acquisition and customer retention. First, in terms of customer acquisition, there are certain business categories and business types that are more likely to remain as loyal customers than others. Therefore, ShopKeep should focus more of its efforts on acquiring more loyal types of customers. There are also several other indicators, such as transaction volatility and number of daily transactions, that ShopKeep can observe during each customer's trial period to determine which trial accounts it should spend more time converting. Second, ShopKeep can use the key indicators of customer attrition that we identified to determine the customers who are more likely to attrite and spend more time on retaining these customers. For example, our model indicates that customers who have been subscribed for longer periods of time are actually more likely to attrite. This indicates that ShopKeep must continue to spend efforts engaging its older customers.

2. Problem statement

ShopKeep is a software company that provides point of sales (POS), inventory management, customer service system, and staff management for stores, including restaurant, shoeshop, grocery, barber shop, bars, and etc. The monthly subscription fee for the using the system is $49 per machine. However, not all the customers will always continue using ShopKeep. Some of the customers might stop using ShopKeep's service for whatever reason, and some of them might come back to ShopKeep after they left. In fact, ShopKeep has seen over 40% of its customers attrite since 2012. In order to increase the customer loyalty as well as lower the churn rate, it's important to find out the main factors that could influence the churn rate and target at those factors to improve the retention rate.

2.1 Data and Sources

The data we used came directly from the company. We have included a detailed description of each variable in Exhibit 1. Broadly speaking, there are three primary types of data that we used:

- **Account profile**: This type of data includes information about each unique account such as its business category, the referral source, the number of subscriptions per customer, the ShopKeep contact, its current subscription status, and any previous status changes. Through the status changes, we are able to to identify which customers have stopped using the service. This data is all cross-sectional as it just captures each account's profile at a given point in time.
- **Accounts' transactions**: This type of data includes information about the daily transactions conducted through ShopKeep by each of the accounts including total transaction amount, number of transactions, minimum / maximum transaction size and gratuity amounts. This data is both cross-sectional as well as a time-series because it is essentially the daily transaction amount over time across each of the stores.
- **ShopKeep's revenues**: This information shows the amount of revenue generated to ShopKeep by each account as well as any refunds made to the customer. This data is also both cross-sectional as well as a time series because it captures the monthly transactions between ShopKeep and all of the accounts.

3. Analysis Methodology

In putting together our analysis, we took the following major steps, which are described in greater detail in the sections that follow.

1. **Data pre-processing**: prepared the training data for analysis.

2. **Feature engineering**: used the given data to create additional meaningful variables.
3. **Model development**: tested several predictive models on training data and evaluated their effectiveness.

3.1 Data pre-processing

The raw data provided by ShopKeep posed several issues for direct processing:

1) The numerical data contained characters like '$' which R cannot process as numbers.
2) The raw data are distributed in 6 different files, resulting in 6 datatables when read in R Studio.
3) Some of the data are time-stamped transaction logs, for example, business transactions at each account, and each account's payments to ShopKeep. These data tables contained a large amount of data but provide marginal meaningful input into our model if we use them without pre-processing.
4) Some of the data columns contained 'na' values.
5) There are both categorical data and numerical data in the provided datatables.

In order to overcome these issues, we took the following steps to pre-process the data

1) We used 'gsub' function in R Studio to remove all characters that are contained in numbers, e.g., "$", "%", etc.
2) We combined "tranx" and "tranx_count" datatables, and were able to form a new datatable with the following data for each of the accounts. We merged these columns with the primary "Accounts" datatable.
    a) average daily transaction number
    b) average transaction amount
    c) average transaction gratuity
    d) average daily transaction amount
    e) average daily gratuity amount
    f) average gratuity as a percent of transaction amount (suggesting whether the business receives tips and how large the tips were)

    The mathematically correct way to do this is, for each account on each day, to merge the number of transactions from one table with transaction amounts in the other table. But the 'merge' function in R can only perform datatable combination with one single identifier across two datatables. We attempted to do this by concatenating the Account ID and Date fields into one new column in both tables and perform 'merge' with the newly generated column. In reality, a for-loop to

process over 2 million transactions this way would take an estimated time of more than 10 hours. Therefore, we decided to use a work-around instead.

What we did as an alternative, was to separately process the two datatables. We calculated average daily number of transactions in one table, and average daily total transaction amount and gratuity in the other. Then we merged two datatables using the Account ID as the identifier. This work-around is a very good approximation of the mathematically correct approach, assuming both datatables are a good and large enough random sample of the population.

1) We pre-processed the datatable with status change log. We marked account with records of “paying->past due” or “paying->customer terminated” with a new column “attrite”. The data in this column will be either 1 (attrited) or 0 (not attrited). We merged this column into the “Accounts” datatable. This variable is the target variable that we’re intend to predict.
2) We merged “SK_employee” datatable into the main “Accounts” datatable using “merge” function in R.
3) Some numerical data columns had “NA” values. We used the median of the non-NA values to replace those “NA” values.
4) We classified variables into numerical variables and categorical variables, and binned the categorical variables.

Following the above steps, we were able to combine all datatables into one single “Accounts” spreadsheet, which allowed us to proceed more easily with our models.

3.2 Feature engineering

The data we have so far, including both the raw data and the aggregated data, (e.g., average daily transaction size) are all structural data that describes the static attributes of an account. After discussing with our counterparts at ShopKeep and understanding their business better, we decided to create and add more “behavioral” data points.

The first behavioral data category is each account’s payment pattern. A typical monthly payment is $49 per terminal. However, some of the accounts would pay an amount other than multiples of $49. Additionally, some of the accounts have received refunds, which results in a negative amount in the payment data. The reason behind these two circumstances are unclear to us at individual account level. But these payment pattern suggested irregularities of the account. Thus we created two additional numerical variables:

- **Integer**: number of times an account has paid an amount other than a multiple of $49. There are three possible reasons for this variable to be larger than 0. First, the account might have purchased other product or services from ShopKeep. Second, the account paid taxes in addition to the $49. Third, something irregular has happened. A common scenario is that the account is dormant and is paying a dormant fee. We haven't differentiate the larger-than-zero value of this variable by 3 reasons in this analysis. But we would recommend do so in a more robust analysis.
- **Negative**: number of times an account has received refund. This variable implies that something irregular was going on with the account, which might contribute to customer attrition.

The second behavioral data category is each account's business and transaction situation. Specifically, we aggregated accounts' total revenue (sum of transaction amount and gratuity) on a weekly basis. Then we generated two additional numerical variables:

- **Volatility**: the volatility of each account's weekly total revenue. We suspect that the higher the volatility is, the higher risk the business has, and thus the higher chance that the business will attrite
- **Trend**: whether the account's revenue was increasing or decreasing. We define this data as the % change of the accounts weekly revenue in the last 1/3 of time vs. the first 2/3 of time. For example, if an account is active for the past year, i.e., 52 weeks, the data would be the % change of its weekly revenue in the last 4 months from that in the first 8 months. The data could be positive, meaning an increase in revenue, or negative, meaning a decrease in revenue, or NA, if the account has only been active for less than 4 weeks. We suspect that a business with a declining revenue would have a higher chance to attrite than a business with an increasing revenue.

In total, we created four additional "behavioral" variables for our prediction model.

3.3 Model development

After pre-processing our model and creating new variables, we began to build a predictive model. Using the aggregated "Accounts" data sheet, we randomized the data and split it into three parts: 60% for training data, 20% for validation data and 20% for test data.

Once we had our training data, we ran a logistic regression using all of the available variables. The logistic regression model returned a decent AUC of 0.8666, however upon analyzing the summary of the model, we discovered that there were many variables that had no significant impact on the model. The categorical variable of

referral code had a large number of different categories and resulted in over two hundred additional binary variables once binned. Since none of the binned referral code variables were significant in the model, we chose to drop referral codes from our model. Another categorical variable for employee email resulted in over sixty additional binary variables. Although some of them were statistically significant, this variable is rather difficult to explain and has the potential to skew the model. Moreover, employee email could only help identify customers less likely to attrite given any statistically significant email had a negative coefficient. Since this is not helpful for our given task, we chose to drop employee emails from our model.

Upon analyzing our data further, we realized that we had included several collinear variables. Therefore, we dropped the Daily.Amount and Daily.Gratuity variables because they were already captured by Daily_total and Gratuity. We also dropped Average_transaction_amount and Average_transaction_gratuity for similar reasons. A summary of all aggregated variables showing which were ultimately used is provided in Exhibit 2.

After identifying the variables we would use, we experimented with logistic regression, Naive Bayes, GBM and big data logistic regression models. We ultimately chose to use a simple logistic regression model because it would be the easiest to analyze the key variables. For our project, it is more important to be able to identify the key drivers of customer attrition rather than be able to create a super accurate model to identify customers who will attrite. Our final model was built using a logistic regression model on 3 categorical variables and 11 numerical variables.

4. Results Summary

The final R code (see Appendix 2) we used for building a logistic regression model, resulted in a relatively high AUC of 0.8749091. Given this number is quite high, which shows that our model gives a relatively accurate prediction for the data. Additionally, we identified 17 variables (including binned variables) across 10 different attributes that showed chances of not being statistically significant of less than 1%, including:

Category	Variable	Significance level	coefficient	Variable Description
ACCOUNTS.Business.Category	Retail	***	-5.792e-01	Retail category accounts
ACCOUNTS.Business	Clothing. Accessories	***	7.604e-01	Clothing or Accessories type accounts

.Type	Recreation Entertainment	**	7.486e-01	Recreation Entertainment type accounts
	Specialty Services	***	7.165e-01	Specialty Services type accounts
	Specialty Food	**	7.255e-01	Specialty Food type accounts
	Specialty Retail	**	6.457e-01	Specialty Retail type accounts
TRANSACTIONS.DAILY.Count.Of.Transactions		***	-1.337e-03	Average number of daily transactions per account
Integer		***	-5.944e-02	# of times an account has paid amounts other than a multiple of $49
Negative		***	7.358e-01	# of times an account has received a refund
Change		***	-3.324e-05	% of change in average weekly transaction amount in the last 1/3 weeks divided by average weekly transaction amount in the first 2/3 weeks
Weeks_Subscribed		***	5.736e-02	# weeks the customer has been a subscribed customer
Active		***	-3.245e-03	# of weeks active divided by # of weeks subscribed
Volatility		**	1.515-03	Standard deviation of weekly transaction amount divided by average weekly transaction amount
Lifecycle.First.Subscri	Disabled	***	3.895e+00	Before subscription, the customer status is labeled as disabled

bed.Status. Before Event	Past.due	***	3.261e+00	Before subscription, the customer status is labeled as Past due
	Past.trial	***	3.299e+00	Before subscription, the customer status is labeled as past trial
	Trial	***	2.934e+00	Before subscription, the customer status is labeled as Trial

5. Principle findings, recommendations and conclusion

5.1 Model interpretation

Using data from the table in the previous section, we can see the model tells us that:

- **Business category data**: The business category data is meaningful primarily for accounts in the retail category. The negative coefficient for retail customers means retail customer tends to be more likely to stay.
- **Business type**: The business type data is meaningful in the sense that customers with certain business types tend to be more likely to attrite. This is evidenced by the fact that the coefficient for several business types, such as Clothing and Accessories, is positive. This means that these customers tend to be less likely to continue to subscribe to ShopKeep's services.
- **Daily counts of transactions**: The average number of daily transactions is significant and has a negative coefficient. This means that the more frequently a certain customer uses the service, the more likely it is for this customer to remain a subscribing user.
- **Integer**: the Integer is significant with coefficient of negative. This indicates that customers who have paid non-standard fee amounts tend to be less likely to drop out. This may be because the company acquired them through some promotion, and the customer was subsequently quite satisfied with the service.
- **Negative**: The more times a customer has received a refund from the company, the less likely it will be to stay with the company. This is probably because the company charged the wrong fee for service or the customer raised a complaint which led to the issuance of a refund. SInce these customers may be unhappy with some aspect of the service, they are more likely to leave.
- **Change**: This variable represents the usage trend for each customer. The negative coefficient indicates that a higher 'change' variable results in less likelihood of attrition. A higher 'change' results when the average transaction amount has increased more in recent weeks as compared to earlier weeks. It

may imply that the customer's business is flourishing or that it has become more reliant on ShopKeep's services.

- **Weeks subscribed**: This variable is also significant with positive coefficient, which means that the longer a customer using the product, the more likely it will be to drop out. The boxplot in Exhibit 3 clearly shows that customers who have subscribed for over 50 weeks are significantly more likely to attrite.
- **Active**: This variable is significant with negative coefficient, which means that if a customer has been active for a larger percentage of its subscription period, it is more likely to will continue using the service in future.
- **Volatility**: Volatility is significant with a positive coefficient. This is in line with what we expected. The higher the volatility of the transaction amount is, the higher risk the business has, and thus the higher chance that the business will attrite.
- **Lifecycle.First.Subscribed.Status.Before Event**: we wanted to measure if the status of certain customer before subscription will influence the chance of attrition, and the result shows this variable is significant. However, the data also says no matter customer is under whatever type of status, it is more likely to atrite, which means this variable is not useful adding information. Thus we decided to exclude it in our further analysis.

5.2 Prediction

As mentioned above, the prediction result of this model looks pretty good, with an AUC of 0.8749091.

While building the model, we randomized the data and did several rounds of prediction models and the result we get was consistently above 0.85. This shows that the model we built using different training data works with different validation and test data. In the sense, it means this model is reliable and doesn't have any major overfitting issues.

By applying this model, we believe the company will have a reliable way of predicting the future behavior of their customer.

5.3 Recommendations for ShopKeep

Based on the model and the result we get from the model, there are several useful pieces of information we want to provide to the company.

- **Customer acquisition**: Certain segments of customer tend to be more loyal than others, thus it makes sense for company to work harder and to acquire specific groups of potential loyal customer and spend less effort on those with higher likelihood of attrition
- **Customer retention**: For customers that have been identified as likely to attrite, the company may want to pay additional attention to them. For example, offer

promotions or other campaign to increase their stickiness and engagement with the service.

- **Signals for customer attrition**: According to our model, there are several warning signs before certain customers decide to attrite. These signals include a lower average number of transactions, a decreasing amount of service usage recently, issuances of refunds and a higher volatility of business associated with a customer. The company should pay attention for these signals. If it shows up for certain customer, they should think about contacting them to understand what's going on, and trying to make the customer stay.
- **Old customer or new customer**: The weeks subscribed variable with positive coefficient may not make sense at first, since it implies that customers that have used ShopKeep for longer are more likely to attrite. However, if we look at it as customers having a certain chance of attriting for every week they use the service, the chance of attrition increases the longer they subscribe. Therefore, it seems that customers will continuously evaluate ShopKeep's services even after they have used it for a period of time. Therefore, the company must continue to actively engage its older customers, primarily those who have subscribed for over 50 weeks, in order to retain them.

Appendix I: Exhibits

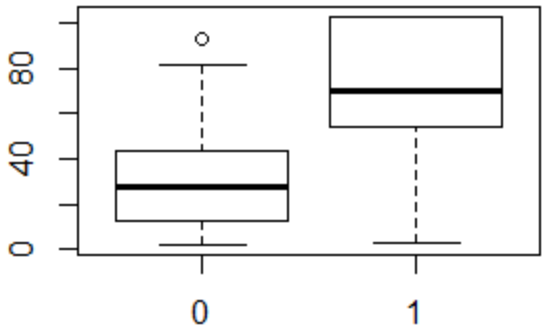
Exhibit 1: Description of data provided by ShopKeep

Raw Data		
File Name	**Variable Description**	**Variable Description**
Revenue	Account.Unid	Each customer's unique account ID
	Charged.Date	Monthly payment date of customers
	Amount	Monthly payment amount of customers
Account	Account.Unid	Each customer has a unique account ID
	Business.Category	Business category of customers
	Business.Type	Business type of customers
	Register.Licenses	# of machines licensed by the customer
	First.Subscribed.Date	Date the customer first began the subscription
	Status.Before.Event	Whether the customer converted from a trial account or after completing a trial account
Status_Change	Account.Unid	Each customer's unique account ID
	Logged.At.Time	The time at which the status change occured
	Status.Change	Trial: one month free trial period Post_trail: Allowed the free trial to lapse after the first one month Paying: Paying customer Customer_terminated: Terminated the service after becoming a paying customer Past_due: Stop paying the license's fee without terminating the service
	Next.Event.At	Time of the subsequent status change
	User.Seq.Number	# of status changes
	Status.Seq.Number	# of times the customer has held the current status
Tranx	Account.Unid	Each customer's unique account ID
	Operation.Date	Operation date
	Amount	Revenue (excluding gratuity) on the given date
	Gratuity	Gratuity amount received on the given date
	Min.Amount	Minimum transaction amount on the given date
	Min.Gratuity	Minimum transaction gratuity on the given date
	Max.Amount	Maximum transaction amount on the given date
	Max.Gratuity	Maximum transaction gratuity on the given date
Tranx_count	Account.Unid	Each customer's unique account ID
	Operation.Date	Operation date
	Count.Of.Transactions	# of transactions on the given date
SK_employee	Account.Unid	Each customer's unique account ID
	Shopkeep.Employee.Email	Account representative's email

Exhibit 2: Description of aggregated data including additional variables

Aggregated Data			
File Name	**Variable Description**	**Variable Description**	**Used?**
Subscribed_account	X	Row number	NA
	Account.Unid	Each customer's unique account ID	NA
	Business.Category	Business category of customers	✓
	Business.Type	Business type of customers	✓
	Referral Code	Referral source of the customer, if any	×
	Register.Licenses	# of machines licensed by the customer	✓
	First.Subscribed.Date	Date the customer first began the subscription	×
	Status.Before.Event	Whether the customer converted from a trial account or after completing a trial account	✓
	Paying Customer Terminated	0: Customers who have not terminated the service 1: Customers who terminated the service	×
	Paying Customer past_due	0: Customers who have not let the payment lapse 1: Customers who have let the payment lapse	×
	Attrite	0: Currently subscribing customer 1: Customers who have either terminated the service or let the payment lapse	NA
	Attrition.Date	Date customer attrited	×
	Shopkeep.employee.email	Account representative's email	×
	Count.Of.Transactions	# of daily transactions	✓
	Daily.Amount	Average daily revenue (excluding gratuity)	×
	Daily.Gratuity	Average daily gratuity amount received	×
	Average_transaction_amount	Average revenue per transaction	×
	Average_transaction_gratuity	Average gratuity per transaction	×
	Daily_total	Sum of daily average revenue and gratuity	✓
	Transaction_total	Sum of average revenue and gratuity per transaction	✓
	Gratuity	Average % of gratuity received per transaction	×
	Integer	Number of times an account has paid amounts other than a multiple of $49	✓
	Negative	Number of times an account has received a refund	✓
	Volatility	Standard deviation of weekly transaction amount divided by average weekly transaction amount	✓
	# of weeks active	# weeks the customer has been active - based on transaction data	✓
	# of weeks subscribed	# weeks the customer has been a subscribed customer	✓
	Active	# of weeks active divided by # of weeks subscribed	✓
	Change	% change in average weekly transaction amount in the last 1/3 weeks divided by average weekly transaction amount in the first 2/3 weeks	✓

Exhibit 3: Weeks subscribed vs. customer attrition



Appendix II: Technical

```
rm(list=ls())

source("http://www.stanford.edu/~bayati/oit367/T367_utilities_11_beta.R")

setwd("C:/Users/lindacguo/Dropbox/ShopKeep")

training = read.csv("subscribed_accounts.csv")

#-----------------------------------------------------------------------------------------#
# Remove NAs from numerical variables.

training$Active = as.numeric(training$Active)
training$Active[is.na(training$Active)] =median(training$Active, na.rm=TRUE)

training$Volatility = as.numeric(training$Volatility)
training$Volatility[is.na(training$Volatility)] =median(training$Volatility, na.rm=TRUE)

training$Weeks_Subscribed = as.numeric(training$Weeks_Subscribed)
training$Weeks_Subscribed[is.na(training$Weeks_Subscribed)]
=median(training$Weeks_Subscribed, na.rm=TRUE)

training$Change = as.numeric(training$Change)
training$Change[is.na(training$Change)] =median(training$Change, na.rm=TRUE)

#-----------------------------------------------------------------------------------------#
# Identifies and separates the categorical and numerical variables.

names.categoricals =
c("ACCOUNTS.Business.Category","ACCOUNTS.Business.Type","LIFECYCLE.FIRST.SUBS
CRIBED.Status.Before.Event")
names.numericals =
c("ACCOUNTS.Register.Licenses","TRANSACTIONS.DAILY.Count.Of.Transactions","daily_t
otal","transaction_total","Integer","Negative","Volatility","Weeks_Active","Weeks_Subscribed","
Active","Change")

predictors = training[,!(colnames(training) %in% c("y"))]
categorical.predictors = predictors[,names.categoricals]
numerical.predictors = predictors[,names.numericals]

#-----------------------------------------------------------------------------------------#
# Converts our categorical variables in categorical.predictors into dummy variables.
```

```
categoricals.binned = bin.cat.data(categorical.predictors)

#-----------------------------------------------------------------------------------------#
# Puts binned categorical predictors, numerical predictors, and the response into a new data
frame

ATTRITE = as.numeric(training$Attrite == "1")
ID=training$X

newPredictors = cbind(categoricals.binned, numerical.predictors)
alldata.binned = as.data.frame(cbind(ID,newPredictors,ATTRITE))

n = nrow(training)
alldata.binned=alldata.binned[sample(n),]
colnames=colnames(alldata.binned)

myVariables = c(colnames)

mytable = alldata.binned[,myVariables]

#-----------------------------------------------------------------------------------------#
# Separate the data set into training, validation and test data

newtraining = mytable[1:(nrow(training)*0.60),]
validation = mytable[(1+nrow(training)*0.60):(nrow(training)*0.80),]
test = mytable[(1+nrow(training)*0.80):nrow(training),]

#-----------------------------------------------------------------------------------------#
# Build logistic regression

model.logreg = buildModel("ATTRITE", predictors=c(), newtraining, type ='classify', method =
'logisticRegression')
pred.logreg.validation = genPred(model.logreg, newdata= validation ,
method="logisticRegression")
pred.logreg.test = genPred(model.logreg, newdata= test , method="logisticRegression")

auc.logreg.validation=auc(validation$ATTRITE,pred.logreg.validation)
auc.logreg.test=auc(test$ATTRITE,pred.logreg.test)

cat("auc log reg = ",auc.logreg.test, " \n")
```